

Constructing a credit curve

The credit curve is as central to the pricing of credit derivatives as the yield curve is to fixed-income derivatives, says David Li. Here, he presents one approach to building it from market information

The structuring and trading of credit derivatives has grown tremendously in recent years. As banks, security firms and corporations continue to hedge and realign their credit risks, and mutual funds strive to enhance their yield and control their credit risk exposure, this growth is expected to continue. The emergence of new credit derivative instruments on Wall Street is now commonplace. In addition to the standard products, such as credit default swaps, total return swaps, credit spread options, collateralised bond obligations and collateralised loan obligations, many new structures incorporating exotic option features have started to appear (for example, the credit default swaption). Many credit derivative products are associated with emerging market sovereign debt: in fact, the volume of emerging market credit derivatives has raised concerns that the total value of underlying assets in emerging market credit derivatives is much larger than that of the actual underlying assets.

How to value these instruments in practice is an interesting, but challenging task. Because most credit derivative transactions are over-the-counter trades, how each transaction is priced is not transparent to outsiders. Different firms may use dramatically different techniques and information in valuing similar instruments. As the credit derivative market develops, the bid and ask spread for typical products shrinks, and standard valuation techniques appear.

Central to the valuation of credit derivatives is the question of how to build a credit curve. A credit curve gives the instantaneous default probability of a party at any time in the future. It is as important to credit derivative pricing as the yield curve is to fixed-income derivative pricing. While many articles have been written about how to build a yield curve based on the current market price information of traded securities – such as futures, bonds and swaps – little has been written on credit curve construction.

This article presents a credit curve construction method based on the yield spread curve of a corporation. We assume that there exists a series of bonds with maturity 1, 2, ..., n years, which are issued by the same company and have the



same seniority. All the information used here is market-observable. From the market price of these bonds, we can calculate their yields to maturity. Using the yield-to-maturity of corresponding treasury bonds, we obtain a yield spread curve. The credit curve construction is based on this yield spread curve and an exogenous assumption about the recovery rate based on the seniority and the rating of the bonds and the industrial sector to which the corporation belongs.

The approach suggested here is contrary to the use of historical default experience information provided by rating agencies such as Moody's Investors Service or Standard & Poor's. We use market information rather than historical information for the following reasons:

- The calculation of profit and loss for a trading desk can only be based on current market information. This current market information reflects the market-agreed perception about the evolution of the market in the future, on which the actual profit and loss depend. The default rate derived from current market information may be considerably different to historical default rates.

- Rating agencies use classification variables in the hope that homogeneous risks will be obtained after classification. This technique has been used elsewhere; for example, in pricing automobile insurance. Unfortunately, classification techniques often omit some firm-specific information.

- Rating agencies react much more slowly than the market in their anticipation of future credit quality. A typical example would be the rating agencies' belated reaction to the recent Asian crisis.

- Ratings are primarily used to calculate default frequency, rather than default severity. However,

much of a credit derivative's value often depends on default severity as well as frequency.

- The information available from a rating agency is usually the one-year default probability for each rating group and the rating migration matrix. Neither the transition matrixes nor the default probabilities are necessarily stable over long periods of time. In addition, many credit derivative products have maturities beyond one year, which requires the use of the long-term marginal default probability.

Of course, using only market information creates problems as well. Except for a few very large companies, the majority do not have a series of outstanding debts with the same seniority and multiple maturities. In the absence of this information, we could still use the asset swap spreads of similar firms as a proxy, or the aggregate spread information of a rating group for the specific company.

When default occurs we assume that the bondholder receives a fixed percentage of the bond price (ie, a recovery rate) immediately prior to default. This is consistent with the approach of Duffie and Singleton (1997). But it differs from the legal treatment of bond default in the US. Under US law, in the case of default on a coupon bond, the bondholder has claim only to the current accrued interest plus the face amount; for a zero coupon bond, the bond holder has claim to a linearly interpolated value of issue price and maturity value based on when default occurs.

The Duffie-Singleton approach provides a particularly good approximation to the legal treatments if the legal treatment is reflected in the market price of the bonds. This default treatment would also simplify modelling in more complicated situations with stochastic hazard rates or interest rates.

In this paper, we consider both the discrete and continuous cases. In the discrete case, the credit curve consists of a series of default probabilities over each period, conditional on survival to the beginning of the period. These probabilities are sometimes referred to as conditional marginal martingale default probabilities (after Jarrow and Turnbull, 1997). In the continuous case, the credit curve can be expressed in terms of a hazard rate func-

tion, which characterises the instantaneous probability of default at any time in the future if it is still "alive". The analysis also demonstrates that under a risk neutral probability measure, the rate of return of risky bonds is not the risk-free interest rate, but rather equals the risk-free rate plus the mean loss due to default (given by the product of the probability of default and the loss given default). It is also shown that the discrete model converges with the continuous one.

The remainder of the paper is organised as follows: in the first section, we describe the characterisation of a default event. Next we define a random variable to denote the survival time of a firm or "time-until-default". The characterisation of time-until-default using a probability density function, an accumulative distribution function, a survival function or a hazard rate function are then presented. Some basic survival notations from actuarial science are used to simplify the analysis. Subsequent sections describe the discrete and continuous models using our framework. The main result in Duffie and Singleton (1997) is reformulated here using our framework. We then give a numerical example and a conclusion.

Time-until-default and the survival function

In the study of default, interest centres on a group of individual companies for each of which there is defined a point event, often called default (or survival), occurring after a length of time. We introduce a random variable called the time-until-default for a security A, for this length of time, and denote it T_A . This random variable is the basic building block for the valuation of cash flows subject to default. To determine time-until-default precisely requires:

- (i) an unambiguously defined time origin;
- (ii) a time scale for measuring the passage of time; and
- (iii) an entirely clear definition of default.

We choose the current time as the time origin to allow use of current market information to build credit curves. The time scale is defined in terms of years for continuous models, or number of periods for discrete models. The meaning of default has been defined by a number of rating agencies, such as Moody's.

Now consider a security A with time-until-default, T_A , defined as a continuous random variable measuring the length of time until default occurs. For notational simplicity we will, for the moment, omit the use of the subscript A. Then let $F(t)$ denote the distribution function of T ,

$$F(t) = \Pr(T \leq t) \quad t \geq 0 \quad (1)$$

and set

$$S(t) = 1 - F(t) = \Pr(T > t) \quad t \geq 0 \quad (2)$$

Assume $F(0)=0$, which implies $S(0)=1$. The function $S(t)$ is called the survival function, and it provides the probability that a security will attain age t . The distribution of T_A can be defined by specifying either the distribution function $F(t)$ or equivalently the survival function $S(t)$. We can also define the probability density function as

$$f(t) = F'(t) = -S'(t) = \lim_{\Delta \rightarrow 0^+} \frac{\Pr[t < T \leq t + \Delta]}{\Delta}$$

To make probability statements about a security having survived x years, we use the notation $T-x|T>x$ to denote its future survival time. In a simple manner we define

$$\begin{aligned} {}_tq_x &= \Pr[T - x \leq t | T > x] \quad t \geq 0 \quad \text{and} \\ {}_tp_x &= 1 - {}_tq_x = \Pr[T - x > t | T > x] \quad t \geq 0 \end{aligned} \quad (3)$$

The symbol ${}_tq_x$ can be interpreted as the conditional probability that the security A will default within the next t years conditional on the survival in x years. In the special case of $x=0$, we have

$${}_tp_0 = S(t) \quad x \geq 0$$

If $t=1$, we use the actuarial convention of omitting the prefix 1 in ${}_tq_x$ and ${}_tp_x$. Thus

$$\begin{aligned} p_x &= \Pr[T - x > 1 | T > x] \\ q_x &= \Pr[T - x \leq 1 | T > x] \end{aligned}$$

The symbol q_x is usually called the marginal default probability, which represents the probability of default in the period of $[x, x+1]$ conditional on the survival to the beginning of the period. A credit curve in a discrete world can be expressed as the sequence of $q_0, q_1, \dots, q_n$.

Hazard rate function

The distribution function $F(t)$ and the survival function $S(t)$ provide two mathematically equivalent ways of specifying the distribution of the random variable of time-until-default. There are, of course, other ways of characterising the problem. A familiar approach used by statisticians is the hazard rate function, which gives the instantaneous default probability for a security that has attained age x .

$$\begin{aligned} \Pr[x < T \leq x + \Delta x | T > x] &= \frac{F(x + \Delta x) - F(x)}{1 - F(x)} \\ &= \frac{f(x)\Delta x}{1 - F(x)} \end{aligned}$$

The hazard rate function

$$h(x) = \frac{f(x)}{1 - F(x)}$$

has a conditional probability density interpolation as the conditional probability density function of default at exact age x , given survival to that time. The relationship between the hazard rate function, the distribution function and the survival function is

$$\begin{aligned} h(x) &= \frac{f(x)}{1 - F(x)} \\ &= -\frac{S'(x)}{S(x)} \end{aligned} \quad (4)$$

The survival function can then be expressed in terms of the hazard rate function using

$$S(t) = e^{-\int_0^t h(s) ds}$$

We can also express ${}_tq_x$ and ${}_tp_x$ in terms of the hazard rate function as follows

$$\begin{aligned} {}_tq_x &= e^{-\int_0^t h(s+x) ds} \\ &= e^{-\int_x^{x+t} h(s) ds} \end{aligned} \quad (5)$$

In addition

$$\begin{aligned} F(t) &= 1 - S(t) \\ &= 1 - e^{-\int_0^t h(s) ds} \end{aligned}$$

and

$$\begin{aligned} f(t) &= S(t) \cdot h(t) \\ &= h(t) \cdot e^{-\int_0^t h(s) ds} \end{aligned}$$

which is simply the density function for T . We need to put a constraint of positiveness on $h(x)$ so that the probability ${}_tp_x$ is less than 1. A typical assumption is that the hazard rate is a constant, h . In this case, the density function is

$$f(t) = he^{-ht} \quad t > 0, h > 0$$

which shows that the density function follows an exponential distribution with parameter h . Under this assumption, the survival probability over the time interval $[0, t]$ is

$$\begin{aligned} {}_tp_x &= e^{-\int_0^t h(x+s) ds} \\ &= e^{-ht} \\ &= (p_x)^t \end{aligned}$$

where p_x is the probability of survival over one year period. This assumption has been implicitly assumed in studies such as Jonkhart (1979) and Yawitz (1977).

The discrete version

Let's begin with the valuation of a corporate bond subject to default. Suppose that a bond promises to pay a stream of cashflows $C_1, C_2, \dots, C_n$ at times $t_1, t_2, \dots, t_n$ in the future. We assume that the current time is t_0 . The discount factor calculated from the Treasury yield curve is $P(t_0, t)$, and is defined as the value of a zero coupon bond at time t_0 which pays \$1 at time t . Denote $V(t_i)$ as the bond's market value at time t_i immediately after the cashflow C_i is paid. Since the present value of the risky bond depends on when default occurs and the recovery value depends on the price just prior to default, its value is path dependent.

We first establish a recursive relationship between the bond price at time t_i and the bond price one period later at time t_{i+1} . The current bond price can then be solved by a recursive formula.

Consider the period $[t_i, t_{i+1}]$. If the bond survives, its terminal value is $C_{i+1} + V(t_{i+1})$; if it defaults (and assuming the Duffie and Singleton's treatment of default) the recoverable value is $R(t_{i+1})[C_{i+1} + V(t_{i+1})]$. In the discrete model we assume that recoverable value is made at the end of the default period $[t_i, t_{i+1}]$, ie, at time t_{i+1} . The conditional survival probability over the period is p_i . So we have the following recursive formula

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$$V(t_i) = \frac{P(t_0, t_{i+1})}{P(t_0, t_i)} [P[C_{i+1} + V(t_{i+1})] + (1 - p_i)R(t_{i+1})[C_{i+1} + V(t_{i+1})]] \\ = \frac{P(t_0, t_{i+1})}{P(t_0, t_i)} \times [p_i + (1 - p_i)R(t_{i+1})][C_{i+1} + V(t_{i+1})], \quad (6)$$

where $i=0, 1, 2, \dots, n-1$, and the boundary condition is

$$V(t_n) = 0$$

By consecutive substitution in (6) we obtain

$$V(t_0) = \sum_{i=1}^n P(t_0, t_i) \left(\prod_{j=0}^{i-1} [p_j + (1 - p_j)R(t_{j+1})] \right) C_i \quad (7)$$

Let

$$DC(t_i) = \left(\prod_{j=0}^{i-1} [p_j + (1 - p_j)R(t_{j+1})] \right) \quad (8)$$

be the credit discount factor, and

$$Q(t_0, t_i) = P(t_0, t_i) DC(t_i)$$

be the credit risk adjusted discount factor, then we have initial bond value

$$V(0) = \sum_{i=1}^n Q(t_0, t_i) \cdot C_i \quad (9)$$

Equation 9 shows that the defaultable bond may still be valued as if the promised cashflows are default free if the discount factor is adjusted for default. In general, the credit risk-adjusted discount factor or the total discount factor is the product of the risk-free discount factor and the pure credit discount factor if the underlying factors affecting default and those affecting the interest rate are independent. To extract the marginal default probability, we can just use expression (7) and the following relationship

Discount factors on June 3, 1998

Date	Discount factor
03-Jun-98	0.9999
03-Jul-98	0.99715
03-Aug-98	0.9943
03-Sep-98	0.99147
03-Oct-98	0.9886
03-Nov-98	0.98577
03-Dec-98	0.9829
03-Jan-99	0.98003
03-Feb-99	0.97714
03-Mar-99	0.97423
03-Apr-99	0.97131
03-May-99	0.96837
03-Jun-99	0.96542
03-Jul-99	0.96246
03-Aug-99	0.95949
03-Sep-99	0.95651
03-Oct-99	0.95353
03-Nov-99	0.95054
03-Dec-99	0.94755
03-Jan-00	0.94456
03-Feb-00	0.94156
03-Mar-00	0.93856
03-Apr-00	0.93556
03-May-00	0.93256
03-Jun-00	0.92956

$$p_i = e^{-\int_{t_i}^{t_{i+1}} h(s) ds}$$

If we assume a piecewise constant hazard rate, that is

$$h(s) = h_i, \quad s \in [t_i, t_{i+1}]$$

then we have

$$p_i = e^{-h_{i+1}(t_{i+1} - t_i)}$$

The continuous model

The continuous version of the Duffie and Singleton model can be obtained from equation (7). Recall from equation (5), we have

$$p_j = e^{-\int_{t_j}^{t_{j+1}} h(s) ds}$$

Then,

$$p_j + (1 - p_j)R(t_{j+1}) \\ = e^{-\int_{t_j}^{t_{j+1}} h(s) ds} + \left(1 - e^{-\int_{t_j}^{t_{j+1}} h(s) ds} \right) R(t_{j+1}) \\ \approx e^{-\left(1 - R(t_{j+1})\right) \int_{t_j}^{t_{j+1}} h(s) ds}$$

if $\Delta t = t_{j+1} - t_j$ is small. Then the credit discount factor can be written as

$$\prod_{j=0}^{i-1} p_j + (1 - p_j)R(t_{j+1}) \\ \approx \prod_{j=0}^{i-1} e^{-\left(1 - R(t_{j+1})\right) \int_{t_j}^{t_{j+1}} h(s) ds} \\ = e^{-\sum_{j=0}^{i-1} \left(1 - R(t_{j+1})\right) \int_{t_j}^{t_{j+1}} h(s) ds}$$

If we divide the time interval $(0, t]$ into many smaller time intervals with a length of $\Delta t = t/n$, then we would have

$$\lim_{n \rightarrow \infty} \left[p_j + (1 - p_j)R \right] \\ = \lim_{n \rightarrow \infty} e^{-\sum_{j=0}^{i-1} \int_{t_j}^{t_{j+1}} (1 - R(t_{j+1})) h(s) ds} \\ = e^{-\int_0^t [1 - R(s)] h(s) ds}$$

If we also assume

$$P(t_0, t_i) = e^{-\int_{t_0}^{t_i} r(s) ds}$$

where $r(s)$ is the short rate, then we can rewrite equation (7) as

$$V(t_0) = \sum_{i=1}^n C_i \cdot e^{-\int_0^{t_i} [r(s) + (1 - R(s)) h(s)] ds} \quad (10)$$

This is the fundamental result of Duffie and Singleton for a zero coupon bond, shown here for a coupon-bearing bond and using only elementary mathematics.

A numerical example

We give a numerical example in this section to illustrate how to build a credit curve. The market information used can be either a series of bond prices or asset swap spreads. We assume a corporation has a series of outstanding bonds with maturities of 1, 2, 3, 4, 5, 7 and 10 years. For simplicity, assume the asset swap spreads of these bonds are known and that each bond is priced at par with semi-annual coupon payments, then the coupon rate can be solved. We assume the following information for the bonds:

Coupon	Maturity	Price
7.89%	1 year	100.00
8.13%	2 year	100.00
8.30%	3 year	100.00
8.40%	4 year	100.00
8.48%	5 year	100.00
8.62%	7 year	100.00
8.78%	10 year	100.00

And if the spread drops to 200 basis points, we have:

Coupon	Spread (bp)	Maturity	Price
10.95%	500	1 year	100.00
11.21%	500	2 year	100.00
11.37%	500	3 year	100.00
11.47%	500	4 year	100.00
11.55%	500	5 year	100.00
11.69%	500	7 year	100.00
11.85%	500	10 year	100.00

The discount curve used is the US dollar Libor curve on June 3, 1998. Discount factors corresponding to coupon payment dates are given in the table at the end of this article.

We also assume that the hazard rate function is a constant from one maturity date to another. Suppose the current time is $t_0=0$, and the maturity dates of bonds are $t_1, t_2, \dots, t_n$. We assume that:

$$h(t) = h_i, \quad t_{i-1} \leq t < t_i$$

For the first bond we have the following valuation formula by using equation 10.

$$100 = \frac{10.95}{2} \cdot e^{-\int_0^{0.5} [r(s) + (1 - R(s)) h(s)] ds} \\ + \left(100 + \frac{10.95}{2} \right) \cdot e^{-\int_0^{1.0} [r(s) + (1 - R(s)) h(s)] ds}$$

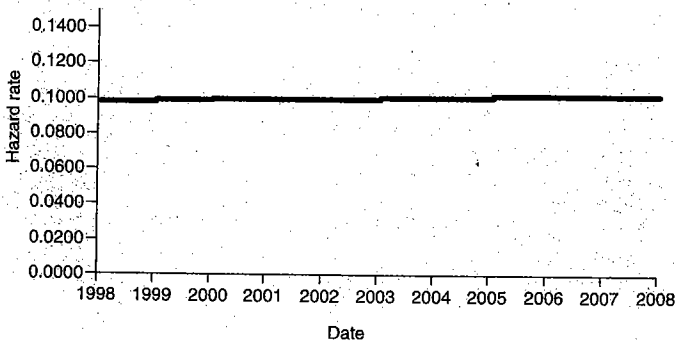
or

$$100 = \frac{10.95}{2} \cdot P(0, 0.5) e^{-(1-R)h_1 \cdot 0.5} \\ + \left(100 + \frac{10.95}{2} \right) \cdot P(0, 1.0) e^{-(1-R)h_1 \cdot 1.0}$$

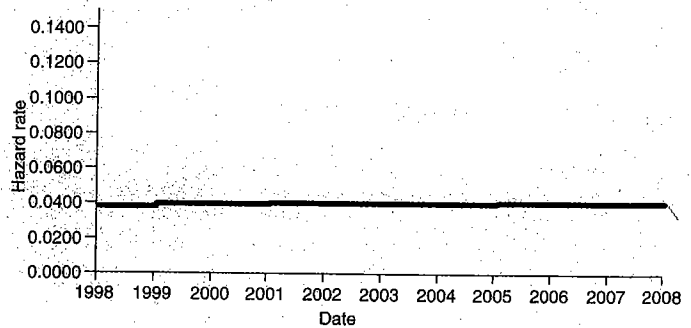
If we assume an exogenous and constant recovery rate R , we can solve for h_1 in the above equation. Having solved for h_1 we can solve for h_2 , so that the second bond maturing at time t_2 is also priced at par. Continuing in this manner

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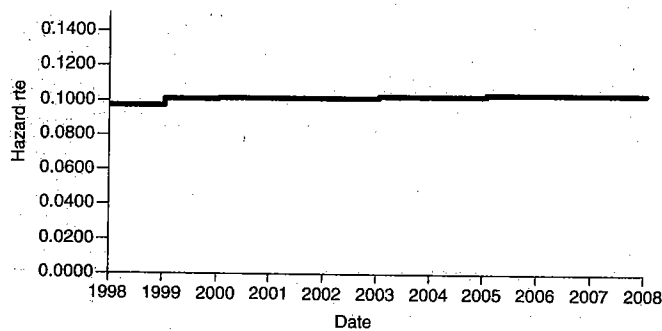
1. Instantaneous default probability spread=500 bp, recovery rate=50%



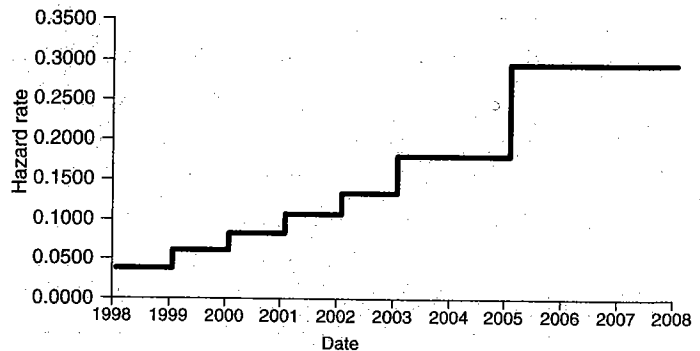
2. Instantaneous default probability spread= 200bp, recovery rate=50%



3. Instantaneous default probability spread= 200bp, recovery rate=80%



4. Instantaneous default probability spread= 200bp in year one and 50bp up per year, recovery rate=50%



we can determine a hazard rate function up to the maturity of the last bond.

Figure 1 gives the hazard rate function when the spread curve is flat at 500bp and the recovery rate is 50%. For a flat spread curve we see that the instantaneous default probability is also flat at 10%. In general, we have $s=(1-R)h$ where s denotes the spread. For a fixed recovery rate, the instantaneous default probability decreases as the spread decreases.

Figure 2 shows that the instantaneous default probability decreases from 10% to 4% when the spread changes from 500bp to 200bp. Also to keep the spread constant, a high recovery rate would result in high default probability.

Figure 3 gives the default probability when the spread is 200bp and the recovery rate is 80%.

The spread curve of a corporation is usually an increasing function of maturity. This reflects the fact that the market values bonds as if corporations have a higher probability of default each year into the future. Litterman and Iben (1991), and Dybvig and Marshall (1996) argue that the market overestimates the risk of long term bonds.

Figure 4 gives the hazard rate function when the spread is an increasing function of maturities. We see clearly that the conditional default probability increases as the maturity increases in this case.

Conclusion

This paper presents a credit curve construction method based on the Duffie and Singleton approach to default settlement. Only observable market information is employed, in a manner similar to yield curve construction. The basic properties of a hazard rate function and its relationship with default probability are presented. We also present an alternative proof of the basic theorem in Duffie and Singleton in both the discrete and continuous case.

Only elementary mathematics are used to prove the discrete case result. The continuous model is shown to be a limiting case of the discrete case. The suggested approach may be used to value many credit derivatives accurately, including credit default swaps and credit linked notes. ■

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The views expressed here are those of the author only, and not necessarily those of CIBC/Oppenheimer or any other organisations of CIBC World Markets Company.

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